

FABIO CAMPOS VILLAYERDE Mechanical Engineering Student | 3D Design & Simulation

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EDUCATION

BSc in Mechanical Engineering | University of Vigo, Spain Focus: Design and Manufacturing | Expected Graduation: June 2026

- **Bachelor Thesis (TFG):** Design of cryogenic structures for the international **DUNE experiment**.
- **Relevant Coursework:** Achieved **Outstanding (9.0/10)** in Data Analysis, Simulation & Validation Systems. Strong background in Graphic Expression and Machine Design.

ENGINEERING PROJECTS

DUNE Experiment Cryogenic Sensor Support (Bachelor Thesis)

- Designing a structural support system for **photon detection instrumentation** operating at **cryogenic temperatures (-186°C)** for the **DUNE experiment**.
- Currently in the **conceptual design phase**, conducting research on material properties for Liquid Argon environments and analyzing thermal contraction constraints to define the final assembly.

High-Torque Direct Drive Sim Rig | Mechatronics & Reverse Engineering

- Built a **15Nm Force Feedback** wheel using a repurposed mobility motor and the **FFBeast** open-source ecosystem. Designed custom 3D printed cooling and housing.
- Engineered hydraulic-feel pedals and a drift handbrake using **100kg Load Cells** and elastomers. Integrated a **SimHub** telemetry system with LED matrices and haptic bass shakers.

IoT Smart Climbing Trainer | Electronics & Data Acquisition

- Developed a precision hangboard using **Strain Gauge Load Cells** and an **ESP32** microcontroller to measure finger strength.
- Programmed a Bluetooth interface to transmit data (**50Hz**) to a custom mobile app for real-time force visualization and progress tracking.

Modular Granular Conveyor Belt | Parametric Design (SolidWorks)

- Designed a scalable aggregate transport system using **parametric modeling**. Divided the assembly into functional modules (Drive, Lattice Body, Tensioner) for transport efficiency.
- Solved site-specific length constraints by creating a flexible assembly that adapts automatically without component redesign.

TECHNICAL SKILLS

- **CAD/CAE & Simulation:** **SolidWorks** (Advanced - Parametric Design), **Fusion 360**, **ANSYS** (Learning), **CATIA** (Basic).
- **Engineering:** Reverse Engineering, Additive Manufacturing (3D Printing), Cryogenic Structures, Mechanism Design.
- **Programming & Electronics:** **Python** (Data Analysis), **C++** (Arduino/ESP32), IoT integration, LLM Integration.
- **Languages:** Spanish & Galician (Native), **English (B1 Certificate)**, German (Basic).

EXPERIENCE

Construction & Maintenance Worker | Campos Nuñez S.L.

Summers 2018 – 2023

- Gained practical insight into structural assembly and on-site technical execution in high-pressure environments.

Climbing Coach (Volunteer) | Grupo de Montaña Alud

Ongoing

- Training competitive athletes at the regional level, focusing on discipline and performance planning.